# Forced dynamic reactor operation for boosting the activity of palladium-based methane oxidation catalysts

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## Background

In the context of establishing a sustainable energy system, methane ($CH_4$) is considered an energy carrier of premier relevance since it can be obtained from renewable resources, i.e. biogas or power-to-gas processes operated with (excess) energy from wind and solar. Both stationary applications, e.g. cogeneration plants, and mobile applications, e.g. hybrid electric vehicles (HEVs), can translate the energy bound in $CH_4$ into heat and power. Since these applications typically emit small amounts of methane, highly active catalysts based on palladium (Pd) that can convert the strong greenhouse gas $CH_4$ into less critical carbon dioxide ($CO_2$) and water ($H_2O$) are mandatory.[1] While such catalysts exhibit excellent activity in their fresh state, the competition of surface adsorbates for active sites, aging-induced sintering of noble metal particles, inhibition, and poisoning are challenges to be overcome, in particular in the low-temperature regime. In this respect, the dynamic operation of Pd-based catalysts can significantly promote $CH_4$ conversion both in lean reaction environments as well as under stoichiometric conditions.[2, 3] The present work aims at elucidating how forced dynamic reactor operation (FDO) can account for high methane conversion in gas streams spanning from rich to stoichiometric to lean conditions.

## Results and discussion

Pd-based catalysts supported on $Al_2O_3$, $CeO_2$, and $CeO_2$-$ZrO_2$ were prepared by incipient wetness impregnation and subsequently coated onto monolithic honeycomb substrates. Catalytic activity tests that were performed in a synthetic gas test bench provided a profound set of kinetic data that underscore the enormous potential of boosting the low-temperature pollutant conversion by FDO in terms of activity and long-term performance. For instance, Fig. 1a depicts how dynamic catalyst operation enables complete $CH_4$ conversion over Pd/$CeO_2$ at comparably low temperatures

and with high reproducibility once the catalyst was activated, whereas under static conditions $CH_4$ conversion remained below 40% even at 600°C. The spatially resolved turnover frequency (TOF) data shown in Fig. 1b uncovered that two zones evolve along the monolithic catalyst during dynamic operation: despite the substantially lower reactor temperature of 400°C applied during dynamic operation, the catalytic activity in the front zone was similar to the activity found in static operation at 620°C, and the rear zone was found to be extraordinarily active. In addition to the removal of hydroxyl species during short rich periods that would inhibit $CH_4$ oxidation in lean gas streams, oxidation state gradients evolving during FDO that were uncovered by X-ray photoelectron spectroscopy (XPS) account for the activity boost. Transferring such encouraging yet fundamental findings to complex realistic gas mixtures and demanding high flow rates is possible, which is underscored by the data shown in Fig. 1c: High-frequency dithering, i.e. lean-rich cycling around stoichiometric conditions, can substantially enhance $CH_4$ conversion compared to static operation especially in the low-temperature regime, which can be exploited for emission reduction from HEVs operated on (sustainable) PtX-methane or biogas. The present work highlights the enormous potential of FDO for boosting the activity of heterogeneous catalysts applied in the field of emission control and beyond.

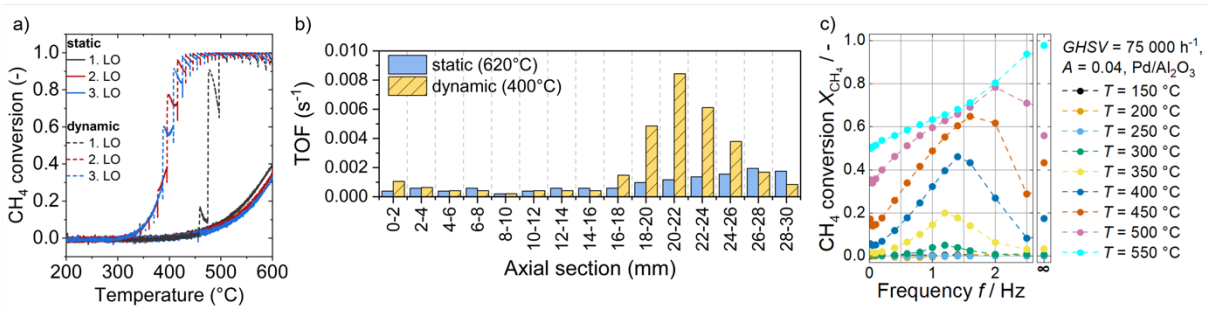


*Figure 1.* a) Light-off in static and dynamic operation of $Pd/CeO_2$ (3200 ppm $CH_4$, 10% $O_2$, 10% $H_2O$ in $N_2$; $O_2$ cut-off every 5 min for 10 s in dynamic operation) and b) turnover frequency (TOF) along the monolithic sample during static and dynamic operation. c) $CH_4$ conversion as function of the dithering frequency over $Pd/Al_2O_3$ operated in a complex realistic gas mixture characteristic for HEV emissions.


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## References


[1] P. Lott, M. Casapu, J.-D. Grunwaldt, O. Deutscmann, *Appl. Catal. B* **2024**, 340, 123241.

[2] D. Hodonj, B. Thiele, O. Deutschmann, P. Lott, *Chem. Eng. J.* **2024**, 499, 155852.

[3] K. Keller, D. Hodonj, L. Zeh, L. Caulfield, E. Sauter, C. Wöll, O. Deutschmann, P. Lott, *Catal. Sci. Technol.* **2024**, 14, 4142-4153.